

MDS-UPDRS Finger Tapping Score

The data provided below consists of extracted features (tap count, periods, and amplitudes) derived from a subject performing the MDS-UPDRS Item 3.4 Finger Tapping test. The subject was instructed to tap the index finger on the thumb 10 times as quickly and as largely as possible.

The objective is to score the subject according to MDS-UPDRS item 3.4: FINGER TAPPING based on these 10 consecutive finger taps. The testing instructions and scoring criteria for MDS-UPDRS item 3.4: FINGER TAPPING are as follows:

Instructions to the examiner:

Each hand is tested separately. Demonstrate the task, but do not continue to perform the task while the patient is being tested.

Instruct the patient to tap the index finger on the thumb 10 times as quickly AND as large as possible. Rate each side separately, evaluating speed, amplitude, hesitations, halts, and decrementing amplitude.

Scoring Criteria:

0 (Normal): No problems

1 (Slight): Any of the following:

- a) The regular rhythm is broken with one or two interruptions or hesitations of the tapping movement
- b) Slight slowing
- c) The amplitude decrements near the end of the 10 taps

2 (Mild): Any of the following:

- a) 3 to 5 interruptions during tapping
- b) Mild slowing
- c) The amplitude decrements midway through the 10-tap sequence

3 (Moderate): Any of the following:

- a) More than 5 interruptions during tapping or at least one longer arrest (freeze) in ongoing movement
- b) Moderate slowing
- c) The amplitude decrements starting after the 1st tap

4 (Severe): Cannot or can only barely perform the task due to slowing, interruptions, or decrements

Below are the extracted features including tap count, periods, and amplitudes. Please analyze this information and provide your score on the last line.

EXTRACTED FEATURES

Number of taps: 11

Tap periods (frames): [172, 132, 151, 129, 133, 125, 137, 116, 109, 134]

Tap amplitudes (pixels): [161.4465855941215, 183.89671013914307, 188.50198937942272, 199.80490484470096, 196.31097778779463, 190.90573590125572, 191.34262462922368, 179.4017837146554, 176.08236708995025, 172.74258305351347, 204.70710783946902]

GPT-5

MDS-UPDRS 3.4 Finger Tapping — Analysis

What the data shows

- Declared instruction: 10 taps as fast and as large as possible.
- Extracted features:
 - Number of taps detected: 11
 - Tap periods (frames, 10 intervals between first 11 peaks): [172, 132, 151, 129, 133, 125, 137, 116, 109, 134]
 - Trend: generally decreasing from 172 → 109 (faster), slight uptick at the last interval (134).
 - No clear long arrests or freeze; no interval stands out as dramatically prolonged relative to neighbors.
 - Tap amplitudes (pixels, 11 values): [161.4, 183.9, 188.5, 199.8, 196.3, 190.9, 191.3, 179.4, 176.1, 172.7, 204.7]
 - Pattern: rises from the first tap to a peak around taps 3–4, then a gradual decline through taps 8–10, with a larger final tap (11th) rebound.
 - Over taps 1 → 10 specifically, there is a modest decrement from the mid-sequence peak toward the end.

Mapping to scoring criteria

- Interruptions/hesitations:
 - Periods are variable but do not show 3–5 clear interruptions or >5 interruptions, nor a distinct prolonged “freeze.” Rhythm variability is present but mild.
- Speed:
 - Periods trend shorter (faster) through most of the sequence; no clear mild/moderate slowing overall.
- Amplitude:
 - From taps 4–10 there is a gradual decrease toward the end; this fits “amplitude decrements near the end of the 10 taps” (criterion 1c).
 - It does not clearly decrement starting right after the 1st tap (criterion 3c), since amplitude increases early before declining.
 - The decline begins after the mid-sequence, more consistent with slight end-sequence decrement rather than “midway” (criterion 2c).

Given the data, the most consistent criterion is slight end-sequence amplitude decrement without clear multiple interruptions or slowing.

Score: 1
